

GENERAL EMBODIMENTS AND VARIATIONS

The embodiments described herein are illustrative and not restrictive. Any feature, structure, step, or component described in connection with one embodiment may be combined with features, structures, steps, or components of other embodiments. The scope of this disclosure includes all substitutions, modifications, and equivalents that fall within the spirit of the described systems and methods.

A. Node Form Factors

Any node may be implemented in multiple form factors. External nodes may be implemented as partially open rings, fully enclosed rings, bands, sleeves, clips, or other wearable forms. Internal nodes may be implemented as intravaginal, anal, or other intraluminal inserts and may include soft overmolded bodies, rigid internal frames, adjustable geometries, or hybrid structures. Context nodes may be implemented as chest-mounted devices, rings, wrist-worn devices, adhesive patches, or other wearable formats.

B. Sensor Modalities and Substitution

Sensors described herein are non-limiting examples. Any node may include one or more modalities comprising pressure sensing, strain sensing, piezoresistive sensing, capacitive sensing, impedance sensing, multi-frequency impedance sensing, temperature sensing, optical sensing including PPG, magnetic sensing including hall-effect sensing and magnets, motion sensing including inertial sensing, electrodermal sensing, pH sensing including ISFET-based sensing, and combinations thereof. Sensor modalities may be substituted, added, removed, duplicated, distributed, or repositioned while maintaining the functions described herein.

C. Controlled Stimulus and Mechanical Variation

Where controlled stimulus is applied, the stimulus may be generated using manual pneumatic actuation, compressible handle elements, micro-pumps, valves, reservoirs, annular cuffs, microbladder rings, constraint bands, mechanical cams, springs, shape-memory elements, or combinations thereof. Stimulus magnitude may be measured using pressure transducers, strain gauges, or other sensors, and may be controlled using closed-loop or open-loop control.

D. Communication and Data Storage

Nodes may communicate using Bluetooth, Bluetooth Low Energy, Wi-Fi, NFC, wired connections, or other communication methods. Nodes may store data locally and upload periodically. The computing system may store data locally, in a remote server, in a cloud environment, or any combination. In some embodiments, data is encrypted at rest and in transit.

E. Processing Distribution and Model Types

Signal processing may be performed within node firmware, on a mobile device, on a remote server, or distributed across components. The system may use rules-based processing, statistical models, machine learning models, deep learning models, personalized models, population-referenced models, and combinations thereof. Models may be updated over time and may incorporate calibration or baseline acquisition routines.

F. Output Formats and Privacy

Outputs may be presented as abstract indices, ratios, zones, ranges, curves, trend lines, distributions, confidence indicators, and/or comparative views. The system may avoid explicit anatomical imagery and explicit anatomical labeling. Outputs may be configured to emphasize change over time, stability, variability, and correlation rather than explicit representations of anatomy.

G. Pairing Contexts and Cross-User Correlation

Cross-user correlation may be performed across different pairing contexts including male-to-female, male-to-male, female-to-female, and other pairings. Correlation may be performed using privacy-preserving metrics and may output alignment, synchrony, stability, confidence, and/or compatibility-style scores without requiring explicit anatomical imagery or labeling.